

Functional Annotation of *paaE* (Q88HS9, PP_3274) — Ring 1,2-Phenylacetyl-CoA Epoxidase, Reductase Subunit — in *Pseudomonas putida* KT2440

0. Identity verification (target confirmation)

Field	Value
UniProt accession	Q88HS9
Gene symbol	<i>paaE</i>
Ordered locus	PP_3274
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125), strain PSEPK
Annotation	Ring 1,2-phenylacetyl-CoA epoxidase, reductase subunit; EC 1.14.13.149
Key domains	Plant-type [2Fe-2S] ferredoxin (IPR001041/IPR006058/IPR036010); β -grasp/NAD-binding scaffold (IPR012675); Cbr1-like FAD-binding oxidoreductase domain (IPR008333)

The identification is unambiguous and internally consistent. The gene symbol *paaE* is the standard name for the reductase subunit of the phenylacetyl-CoA (PA-CoA) oxygenase across bacteria (*E. coli*, *Pseudomonas*, *Azoarcus*), and the *paa* gene cluster is conserved. The three cofactor-binding modules encoded by Q88HS9 — an FAD-binding domain, an NAD(P)H/ β -grasp module, and a plant-type [2Fe-2S] ferredoxin domain — are the textbook architecture of a **flavo-ferredoxin NAD(P)H:acceptor oxidoreductase (reductase)**, the same fold family used by the reductase components of bacterial multicomponent oxygenases. This matches the UniProt description (reductase subunit; EC 1.14.13.149) exactly. No conflicting gene named *paaE* with a different function was found. The report below therefore proceeds with high confidence.

1. Summary (answer to the research question)

paaE encodes the **reductase (electron-transfer) subunit of the multicomponent ring-1,2-phenylacetyl-CoA epoxidase (oxygenase)** that carries out the single committed oxygen-dependent step of the aerobic "hybrid" phenylacetate/phenylalanine catabolic pathway in *P. putida* KT2440. PaaE is a soluble, **cytoplasmic** NAD(P)H-dependent oxidoreductase that uses its FAD and [2Fe-2S] cofactors to shuttle reducing equivalents from **NAD(P)H → FAD → [2Fe-2S] cluster → the diiron active site of the PaaA/PaaC oxygenase core**. This electron supply enables the oxygenase to activate molecular O₂ and **epoxidize the aromatic ring of phenylacetyl-CoA to a non-aromatic ring-1,2-epoxide** (EC 1.14.13.149), which subsequently isomerizes to oxepin-CoA and is degraded to acetyl-CoA and succinyl-CoA. Its substrate specificity is defined by the complex: the oxygenized substrate is the **CoA thioester of phenylacetate (phenylacetyl-CoA)**, while PaaE's own redox substrate is **NAD(P)H**.

2. Biochemical function — what reaction is catalyzed, and substrate specificity

2.1 The enzyme complex and PaaE's specific role

The oxygenation of the aromatic ring in this pathway is performed not by a single protein but by a **multicomponent oxygenase encoded by *paaABCDE***. Biochemical dissection of the homologous *E. coli* system established the roles of the subunits:

- **PaaA** — catalytic α-subunit bearing the **carboxylate-bridged diiron center** where O₂ is activated and the substrate is oxidized.
- **PaaC** — structural β-subunit; PaaA and PaaC together form the catalytic heterotetrameric core (PaaAC).
- **PaaB** — small accessory subunit required for activity.
- **PaaE** — the **reductase**: an NAD(P)H:oxidoreductase carrying FAD and a [2Fe-2S] cluster that feeds electrons into the diiron oxygenase.
- **PaaD** — a putative [Fe-S] chaperone/assembly factor; **not essential** for in vitro catalysis.

In vitro reconstitution showed that "**each of the PaaA, B, C, and E subunits are necessary for catalysis, whereas PaaD is not essential**" (Grishin et al. 2011). Thus PaaE is an obligatory component: without the reductase, the diiron center cannot be re-reduced between turnovers and no oxygenation occurs.

2.2 Overall reaction (EC 1.14.13.149)

The complex catalyzes:



PaaE's partial reaction is the **oxidation of NAD(P)H and delivery of two electrons** (one at a time, via FAD semiquinone chemistry and the one-electron [2Fe-2S] carrier) to the oxygenase. This is the classic division of labour in bacterial multicomponent monooxygenase/dioxygenase systems: a flavo-ferredoxin **reductase** (PaaE) + a **diiron oxygenase** (PaaA/PaaC) + accessory subunit(s).

2.3 Substrate specificity

- **Oxidized substrate of the complex:** the **coenzyme A thioester of phenylacetate** (phenylacetyl-CoA), not free phenylacetate. Intermediates in this entire pathway are handled as CoA thioesters, a defining feature that distinguishes it from classical aerobic aromatic catabolism (Ismail et al. 2003; Teufel et al. 2010). This CoA-activation is imposed upstream by the phenylacetate-CoA ligase (PaaK/PaaF-type ligase).
- **Redox cosubstrate of PaaE:** **NAD(P)H** (the annotated EC uses NADPH); electron acceptor is molecular O₂ at the oxygenase diiron site.
- **Product:** a **ring-1,2-epoxide of phenylacetyl-CoA** — a reactive, *non-aromatic* epoxide. This is mechanistically distinct from ring-hydroxylating dioxygenases, which make cis-dihydrodiols/catechols. As Ismail et al. put it, the pathway generates "a hydroxylated and reduced derivative of phenylacetyl-CoA, which is **not re-oxidized to a dihydroxylated aromatic intermediate**, as in other known aromatic pathways."

2.4 Structural/mechanistic evidence for the electron-transfer role

The crystal structure of the PaaAC catalytic core (*E. coli*) revealed a monooxygenase fold organized **"very differently from other known multisubunit monooxygenases"** and, importantly, that it **"lacks their conservative network of hydrogen bonds between the di-iron center and protein surface, suggesting different association with the reductase and different mechanisms of electron transport"** (Grishin et al. 2011). This directly implicates a dedicated reductase (PaaE) whose docking and electron-relay to the diiron center is a distinguishing property of the Paa oxygenase subfamily. The reductase's own cofactor set — **FAD (2-electron/NAD(P)H interface) + [2Fe-2S] (1-electron output)** — is exactly what is required to interconvert the two-electron chemistry of NAD(P)H and the one-electron chemistry of the diiron/O₂ system.

3. Pathway context — where PaaE functions biochemically

PaaE performs the **committed, rate-defining oxygenation step** of the widespread aerobic "hybrid" phenylacetate pathway (the *phenylacetyl-CoA catabolon*). The full route (Teufel et al. 2010; Ismail et al. 2003; Ferrández et al. 1998; Teufel et al. 2011):

1. **Activation:** phenylacetate + CoA + ATP → **phenylacetyl-CoA** (phenylacetate-CoA ligase; PaaK/PaaF).
2. **Ring epoxidation (PaaE step):** phenylacetyl-CoA → **ring-1,2-epoxyphenylacetyl-CoA** by the *paaABCDE* oxygenase, with PaaE supplying electrons from NAD(P)H. (*EC 1.14.13.149*)
3. **Isomerization:** the non-aromatic epoxide → **oxepin-CoA** (a seven-membered O-heterocyclic enol ether) by the isomerase (PaaG-type).
4. **Ring hydrolysis:** oxepin-CoA → 3-oxo-5,6-dehydrosuberil-CoA by the bifunctional **PaaZ** (enoyl-CoA hydratase + aldehyde dehydrogenase) (Teufel et al. 2011).
5. **β-oxidation-like steps** (PaaF/G/H/J) → **acetyl-CoA + succinyl-CoA**, feeding the TCA cycle.

This pathway is a **"hybrid" aerobic route**: it requires O₂ (via PaaABCDE) yet, unusually, does *not* use an oxygenase for ring *cleavage* — the ring is opened hydrolytically (Fuchs 2008). PaaE is the electron-delivery engine that makes the one oxygen-dependent step possible. Because phenylalanine and styrene are funneled to phenylacetate, PaaE also lies on the terminal common route of **phenylalanine and styrene catabolism**. This pathway operates in **"16% of all bacteria whose genome has been sequenced, including *Escherichia coli* and *Pseudomonas putida*"** (Teufel et al. 2010), placing PaaE firmly in *P. putida* KT2440 physiology.

Regulation/genomic context: the *paaABCDE* genes are part of a large *paa* operon/cluster; in *E. coli* the operon is induced by phenylacetyl-CoA (the true inducer) via the PaaX repressor (Ferrández et al. 1998). The same catabolon logic — CoA-thioester induction and a conserved gene cluster — applies in *Pseudomonas*.

4. Subcellular localization

PaaE is a **soluble cytoplasmic protein**. Multiple lines of evidence support this: - The entire phenylacetate/phenylalanine hybrid pathway processes **intracellular CoA thioesters** (Teufel et al. 2010; Ismail et al. 2003); CoA-thioester metabolism is inherently cytoplasmic because CoA does not cross the membrane, and its substrate is generated by a cytoplasmic CoA ligase. - PaaE was expressed and reconstituted as a soluble subunit in vitro together with the other Paa subunits (Grishin et al. 2011), consistent with a cytosolic, non-membrane enzyme. - The domain architecture (FAD/NAD(P)H/[2Fe-2S] reductase) carries **no signal peptide or transmembrane segment**, matching a cytoplasmic redox protein.

Functionally, PaaE acts at the **cytoplasmic face**, transiently docking onto the PaaA/PaaC oxygenase to transfer electrons.

5. Evolutionary / bioinformatic inference

- **Domain-based inference:** the FAD-binding (Cbr1/ferredoxin-NADP⁺-reductase-like), NAD-binding β-grasp, and plant-type [2Fe-2S] ferredoxin domains define a **reductase of the class-I(A)/glutathione-reductase-like flavo-ferredoxin family** that partners diiron/Rieske oxygenases. This places PaaE alongside well-characterized reductase components of bacterial multicomponent oxygenases and is fully consistent with the assigned reductase role.
- **Orthology/conservation:** *paaE* orthologs co-occur with *paaABCD* in a conserved cluster across diverse phyla (*E. coli paaABCDEFGH*; *Azoarcus evansii paaYZGHIKABCDE*; *P. putida paa*), and their gene products are mutually "similar to the characterized putative gene products involved in PA catabolism in *E. coli* and *Pseudomonas putida*" (Mohamed et al. 2002). Conserved synteny with the oxygenase genes is strong evidence that PaaE is the dedicated reductase of this specific oxygenase.

6. Supported vs. refuted hypotheses

Supported - H1 (supported): PaaE is the **reductase subunit** of the PA-CoA epoxidase and is **essential** for oxygenase activity (Grishin et al. 2011; Ismail et al. 2003; Ferrández et al. 1998). - H2 (supported): The complex epoxidizes the **aromatic ring of phenylacetyl-CoA** to a non-aromatic ring-1,2-epoxide using O₂ and NAD(P)H (Teufel et al. 2010; EC 1.14.13.149). - H3 (supported): PaaE functions **in the cytoplasm** as part of a CoA-thioester-based pathway (Teufel et al. 2010; Ismail et al. 2003). - H4 (supported): The pathway is a **hybrid aerobic route** (O₂ used for ring activation, not ring cleavage) that yields acetyl-CoA + succinyl-CoA (Fuchs 2008; Teufel et al. 2011).

Refuted / excluded - Not a ring-hydroxylating dioxygenase producing catechols/dihydrodiols — the product is an epoxide that isomerizes to oxepin-CoA, and no dihydroxy-aromatic is formed (Ismail et al. 2003; Teufel et al. 2010). - PaaE is **not** the non-essential accessory factor — that role belongs to **PaaD** (Grishin et al. 2011).

7. Limitations and future directions

- Most direct biochemical/structural data (subunit reconstitution, PaaAC crystal structure) come from the *E. coli* ortholog; while the *P. putida* KT2440 *paa* cluster is orthologous and the pathway is confirmed present in *P. putida* (Teufel et al. 2010), a dedicated crystal structure or steady-state kinetic characterization of the specific *P. putida* KT2440 PaaE (Q88HS9) protein was not identified.
- The precise cofactor stoichiometry (FAD vs. FMN; measured [2Fe-2S] EPR signature) and NADH-vs-NADPH preference of the *P. putida* enzyme have not, to my knowledge, been individually reported and would be worth measuring.
- A high-resolution PaaE–oxygenase docking model / electron-transfer pathway remains an open structural question flagged by Grishin et al. (2011).

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